

Regression Model

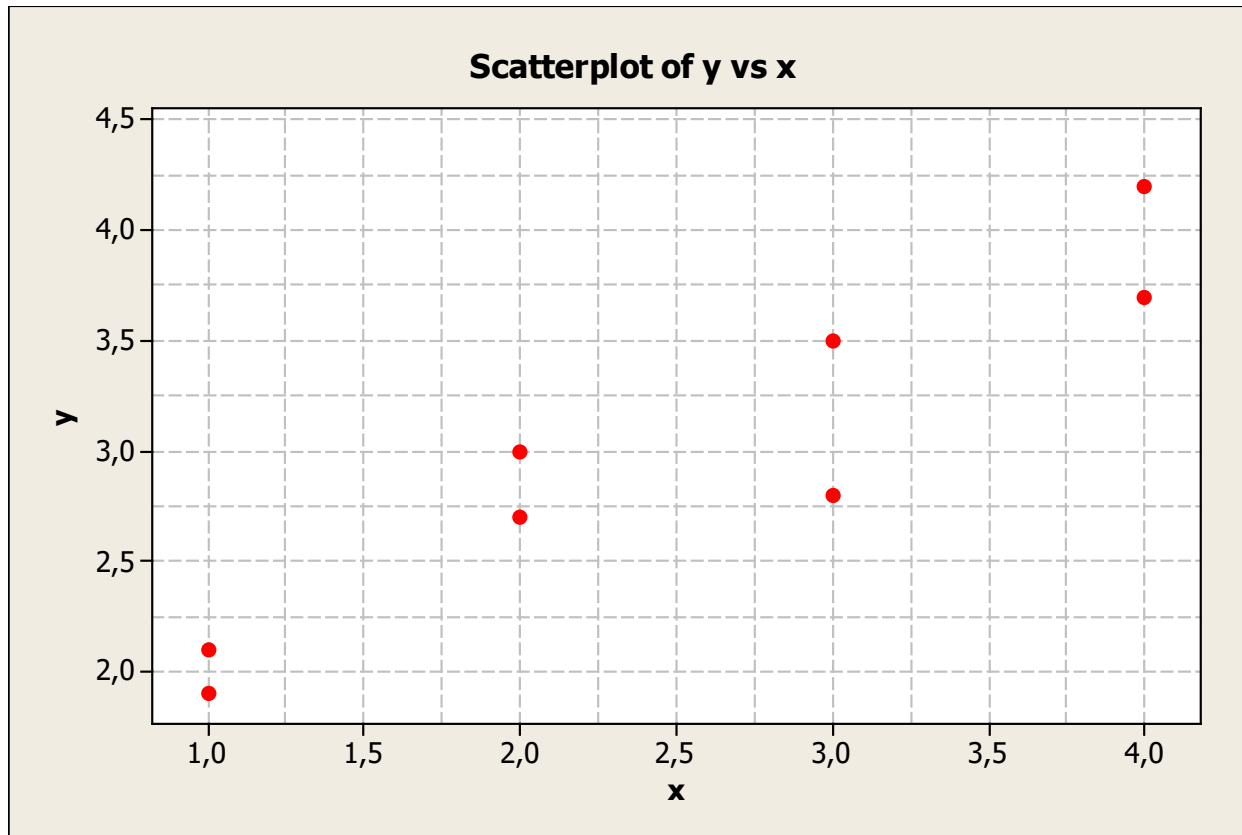
Part B

Expanding the Model: Lack of fit (LOF)

y_{ij} j -th observation at the i -th treatment

$i = 1, 2, \dots, m$ $N = \sum_{i=1}^m n_i$ total number of observations

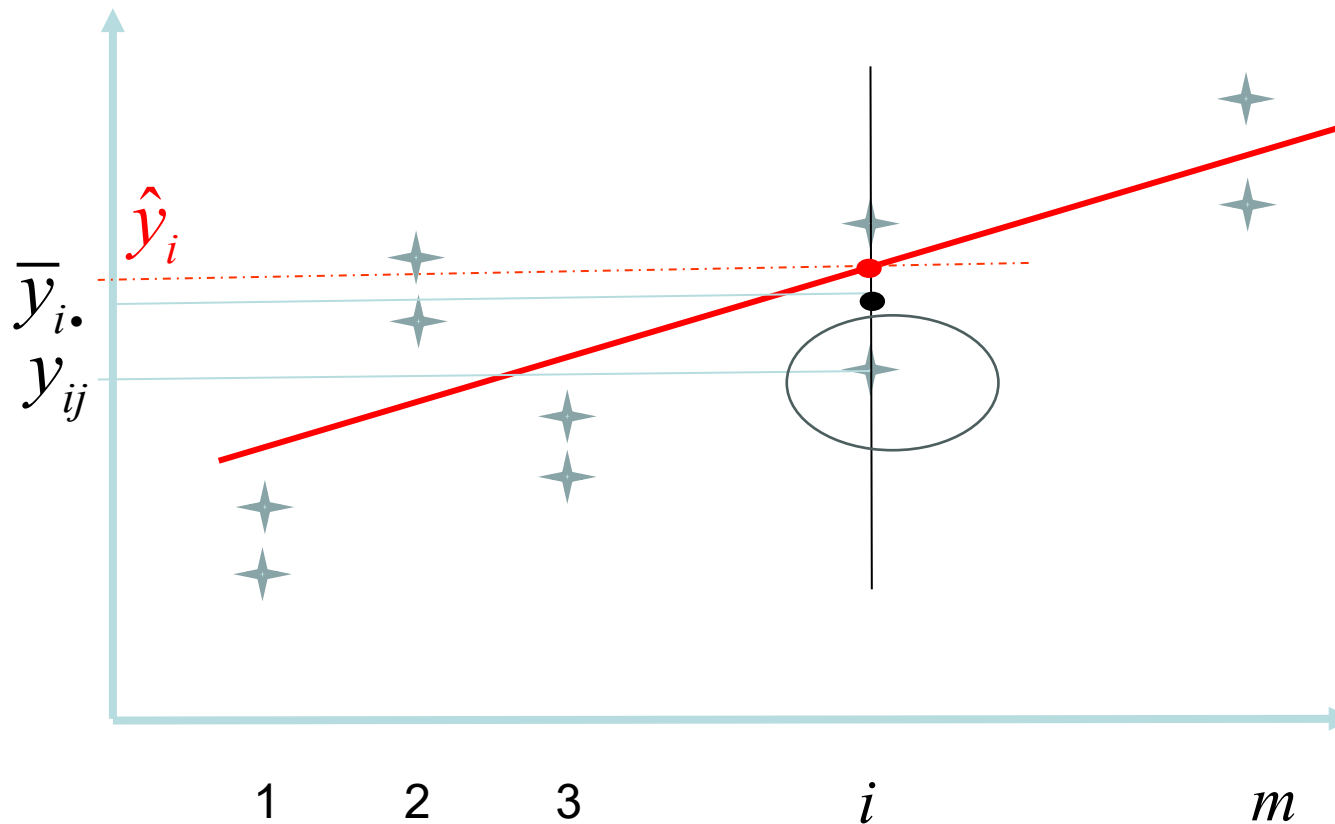
$j = 1, 2, \dots, n_i$



x	y
1	2.1
1	1.9
2	3.0
2	2.7
3	3.5
3	2.8
4	4.2
4	3.7

- Algebraic partition

$$SS_E = SS_{LOF} + SS_{PE}$$



$$e_{ij} = y_{ij} - \hat{y}_i = (y_{ij} - \bar{y}_{i.}) + (\bar{y}_{i.} - \hat{y}_i)$$

$$y_{ij} - \hat{y}_i = (y_{ij} - \bar{y}_{i.}) + (\bar{y}_{i.} - \hat{y}_i)$$

$$\sum_{i,j} (y_{ij} - \hat{y}_i)^2 = \sum_{i,j} (\bar{y}_{i.} - \hat{y}_i)^2 + \sum_{i,j} (y_{ij} - \bar{y}_{i.})^2 =$$

$$= \sum_{i=1,m} n_i (\bar{y}_{i.} - \hat{y}_i)^2 + \sum_{\substack{i=1,m \\ j=1,n_i}} (y_{ij} - \bar{y}_{i.})^2$$

$$SS_E = \sum_{i,j} e_{ij}^2 = \sum_{i,j} (y_{ij} - \hat{y}_i)^2$$

$$df_E = N - p$$

$$SS_{PE} = \sum_{i,j} (y_{ij} - \bar{y}_{i.})^2 = \sum_i (n_i - 1) s_i^2$$

$$df_{PE} = \sum_{i=1,m} (n_i - 1) = N - m$$

$$SS_{LOF} = \sum_i n_i (\bar{y}_{i.} - \hat{y}_i)^2 = SS_E - SS_{PE}$$

$$df_{LOF} = (N - p) - (N - m) = m - p$$

- **Remember:** Replicates allow us to have an estimate of the Pure Error.
- We call it **Pure Error** because it is **NOT** influenced by a possible model distortion (**BIAS**)
- The complete *df* algebraical identity is

$$df_{TOT}^* = df_{REG}^* + df_E = 1 + df_{REG} + df_{LOF} + df_{PE}$$

$$df_{TOT}^* - 1 = df_{TOT} = df_{REG} + df_{LOF} + df_{PE}$$

$$SS_E = SS_{LOF} + SS_{PE}$$

$$N-p = m-p + N-m$$

$$E(MS_{PE}) = \sigma^2$$

$$F_0 = \frac{SS_{LOF} / (m - p)}{SS_{PE} / (N - m)} = \frac{MS_{LOF}}{MS_{PE}}$$

$$E(MS_{LOF}) = \sigma^2 + \frac{1}{m - p} (E(\mathbf{y}) - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{I} - \mathbf{H}) (E(\mathbf{y}) - \mathbf{X}\boldsymbol{\beta})$$

See the supplementary material for the proof

If the model is unbiased

$$E(\mathbf{y}) - \mathbf{X}\boldsymbol{\beta} = \mathbf{0} \quad \longrightarrow \quad \frac{MS_{LOF}}{MS_{PE}} \sim F(m - p, N - m)$$

Analysis of Variance Table

<i>Source</i>	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>p</i>
Regression	1	3.7822	3.7822	41.53	0.001
Residual Error	6	0.5465	0.0911		
Lack of Fit	2	0.1115	0.0557	0.51	0.634
Pure Error	4	0.4350	0.1088		
Total	7	4.3287			

- **Remarks:**

- We test the homogeneity of variances (Bartlett, Brown-Forsythe) if we have replicates
- Heuristic method (not supported in JMP)
- if we do not have a **correct variance estimate**, the test will signal LOF even if we do **NOT** have a **LOF**

Model Bias

Consequences if the model does have a BIAS

Assumed model $\rightarrow \mathbf{y} = \mathbf{X}_1 \boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}$

True model $\rightarrow \mathbf{y} = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_2 \boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$

$$\hat{\boldsymbol{\beta}}_1 = \left(\mathbf{X}_1^T \mathbf{X}_1 \right)^{-1} \mathbf{X}_1^T \mathbf{y}$$

- Properties of the estimates

$$E(\hat{\boldsymbol{\beta}}_1) = \boldsymbol{\beta}_1 + \mathbf{A} \boldsymbol{\beta}_2 \quad \text{con} \quad \mathbf{A} = \left(\mathbf{X}_1^T \mathbf{X}_1 \right)^{-1} \mathbf{X}_1^T \mathbf{X}_2$$

$$V(\hat{\boldsymbol{\beta}}_1) = \left(\mathbf{X}_1^T \mathbf{X}_1 \right)^{-1} \sigma^2$$

Alias Matrix

$$E(\hat{\mathbf{y}}) = E(\mathbf{X}_1 \hat{\boldsymbol{\beta}}_1) = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_1 \mathbf{A} \boldsymbol{\beta}_2 = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{H}_1 \mathbf{X}_2 \boldsymbol{\beta}_2$$

Criticality of data

- Data Criticality comes from:
 - The experiment (the design)
 - The data

Criticality of the Experimental conditions

Some experimental conditions (treatments) may affect negatively the model. How to find them?

$$\mathbf{H} = \begin{bmatrix} h_{ij} \end{bmatrix} \quad \hat{\mathbf{y}} = \mathbf{H}\mathbf{y} \quad \rightarrow \quad \hat{y}_i = \sum_{j=1, N} h_{ij} y_j$$

y_j affects $\hat{y}_i$ through h_{ij} (Leverages)

Usually we focus on the **diagonal elements** of the $\mathbf{H}$

$$\sum_{i=1}^N h_{ii} = \text{Tr}(\mathbf{H}) = \text{Tr}\left(\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T\right) = \text{Rank}(\mathbf{X}) = p$$

If there is **no** particular influence $h_{ii} = \frac{p}{N}$

Critical condition if $h_{ii} \geq 2 \frac{p}{N}$ Minitab warns if $h_{ii} \geq \text{Max}\left(3 \frac{p}{N}, 0.99\right)$

REMARK: You can check it **before** running the experiment

Influence on regression parameters

COOK Coefficient (first order)

$$D_i = \frac{\left(\hat{\mathbf{y}}_{(i)} - \hat{\mathbf{y}}\right)^T \left(\hat{\mathbf{y}}_{(i)} - \hat{\mathbf{y}}\right)}{pMS_E}$$

If the i -th observation does not affect the fits, the coefficient D_i will be close to zero.

We may find other **equivalent expressions**

$$D_i = \frac{\left(\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}}\right)^T \mathbf{X}^T \mathbf{X} \left(\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}}\right)}{pMS_E} = \frac{r_i^2}{p} \frac{V[\hat{y}(\mathbf{x}_i)]}{V(e_i)} = \frac{r_i^2}{p} \frac{h_{ii}}{1 - h_{ii}}$$

If $D_i > 1$ or it is **quite different** from the other coefficients, the related observation is influential

Other coefficients

DFIT (Belsley, Kuh, Welsch)

$$DFIT_i = \sqrt{\frac{(\hat{\beta}_{(i)} - \hat{\beta})^T \mathbf{X}^T \mathbf{X} (\hat{\beta}_{(i)} - \hat{\beta})}{s_{(i)}^2}} = \sqrt{\frac{D_i p MS_E}{s_{(i)}^2}}$$

Minitab has an equivalent formula (with sign)

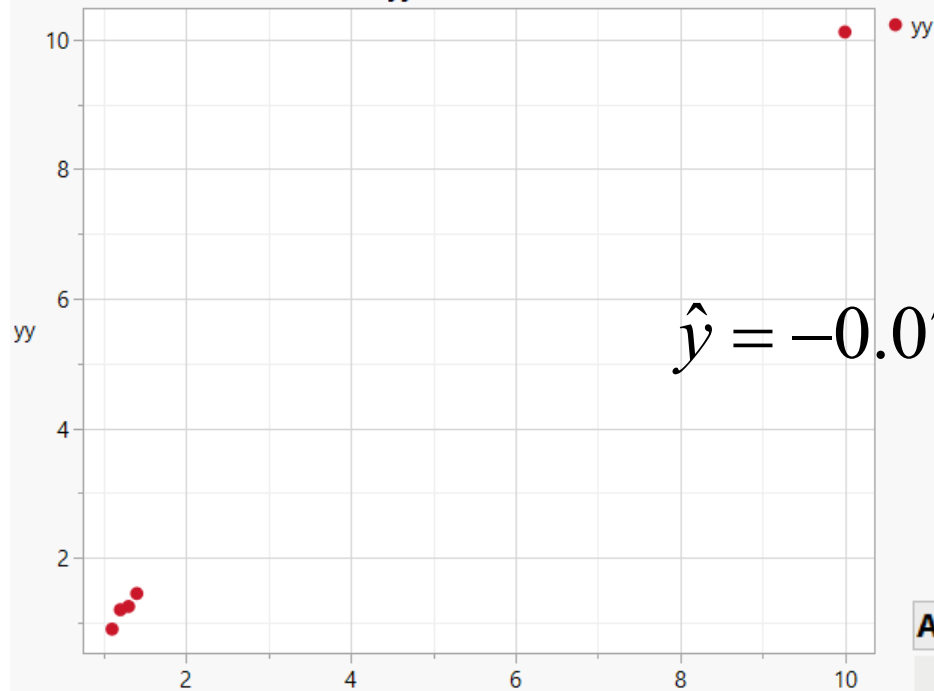
$$DFIT_i = \frac{\hat{y}_i - \hat{y}_{(i)}}{\sqrt{s_{(i)}^2 h_{ii}}} = e_i \sqrt{\frac{N - p - 1}{SS_E (1 - h_{ii}) - e_i^2}} \sqrt{\frac{h_{ii}}{1 - h_{ii}}}$$

Atkinson's Coefficient (useless because is a scaled DFIT)

$$A_i = \sqrt{\frac{(N - p)(\hat{\beta}_{(i)} - \hat{\beta})^T \mathbf{X}^T \mathbf{X} (\hat{\beta}_{(i)} - \hat{\beta})}{p s_{(i)}^2}} = \sqrt{\frac{D_i (N - p) MS_E}{s_{(i)}^2}} = DFIT_i \sqrt{\frac{p}{N - p}}$$

Conclusions: COOK (and DFIT)

yy vs. xx



Example

yy **xx**

0.90 1.1

1.20 1.2

1.25 1.3

1.45 1.4

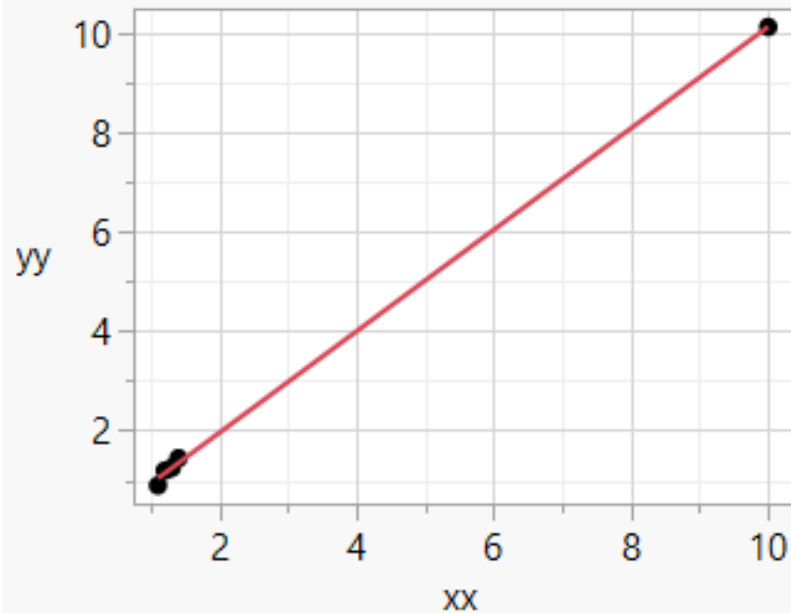
10.12 10.0

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	63.774480	63.7745	5687.382
Error	3	0.033640	0.0112	Prob > F
C. Total	4	63.808120		<.0001*

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob> t	VIF
Intercept	-0.075951	0.062362	-1.22	0.3103	.
xx	1.0199837	0.013525	75.41	<.0001*	1



Fits and Diagnostics for Unusual Observations

Obs	yy	Fit	Resid	Std	Resid
5	10.1200	10.1239	-0.0039	-1.44	X
X	Unusual	X			

RESI1	SRES1	TRES1	HI1	COOK1
0.146031	-1.60190	-3.43919	0.258891	0.45
0.051971	0.56779	0.49071	0.252855	0.05
-0.000028	-0.00030	-0.00025	0.247145	0.00
0.097974	1.06253	1.09854	0.241762	0.18
-0.003886	-1.43653	-2.09943	0.999347	1580.21

Using the model

- We use the model to predict
 - **Point prediction in $\mathbf{x}$** $\hat{y}(\mathbf{x}) = \mathbf{x}^T \hat{\boldsymbol{\beta}}$
 - Confidence Interval of the Expected response (**Confidence Interval - CI**)
 - Prediction Interval of the Point response (**Prediction Interval - PI**)
 - **Confidence Bands** of Expected or Point prediction
- We could use the estimated model in a more complex problem (i.e. optimization problem)

Confidence Interval **CI** (for the expected value)

Let us suppose we are interested in the prediction at point $\mathbf{0}$

Model $y(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\beta} + \varepsilon$

$$E[y(\mathbf{x})] = \mathbf{x}^T \boldsymbol{\beta}$$

Point Estimate $\hat{y}(\mathbf{x}) = \mathbf{x}^T \hat{\boldsymbol{\beta}}$

Let us consider $E[y(\mathbf{x})] - \hat{y}(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\beta} - \mathbf{x}^T \hat{\boldsymbol{\beta}}$

It is a normal random variable (univariate)

$$E[E(y(\mathbf{x})) - \hat{y}(\mathbf{x})] = \mathbf{x}^T \boldsymbol{\beta} - \mathbf{x}^T \boldsymbol{\beta} = 0$$

$$V[E(y(\mathbf{x})) - \hat{y}(\mathbf{x})] = \mathbf{x}^T \sum_{\hat{\boldsymbol{\beta}}} \mathbf{x} = \mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x} \sigma^2$$

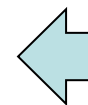
Confidence Interval **CI** (for the expected value)

$$\frac{E[y(\mathbf{x})] - \hat{y}(\mathbf{x})}{\sqrt{\sigma^2 \mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}}} \sim N(0, 1)$$

If we do not know σ^2

$$\frac{E[y(\mathbf{x})] - \hat{y}(\mathbf{x})}{\sqrt{MS_E \mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}}} \sim t(N - p)$$

$$\text{if } \mathbf{x} \equiv \mathbf{x}_i \rightarrow h_{ii} = \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_i$$



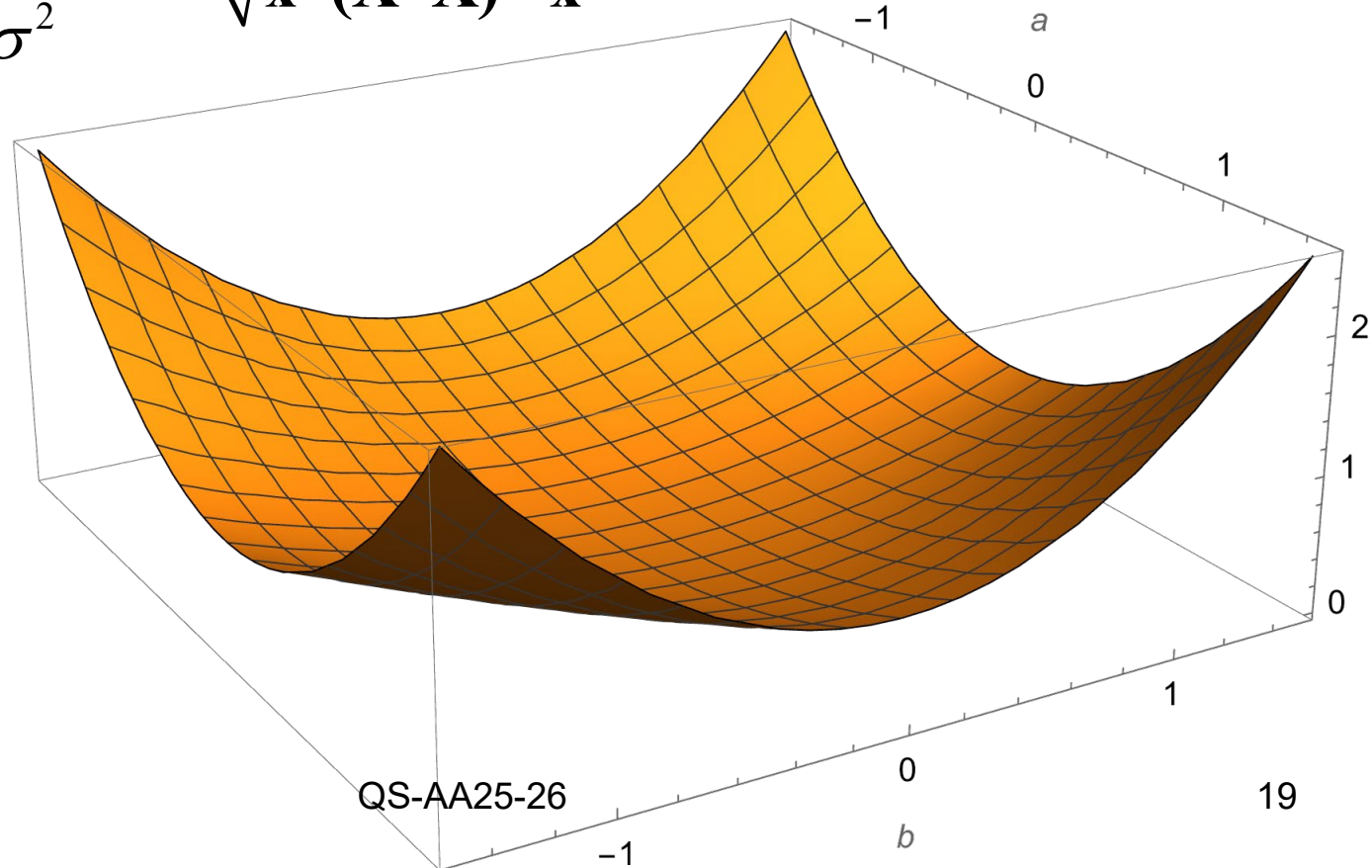
not
constant
Interval

An example

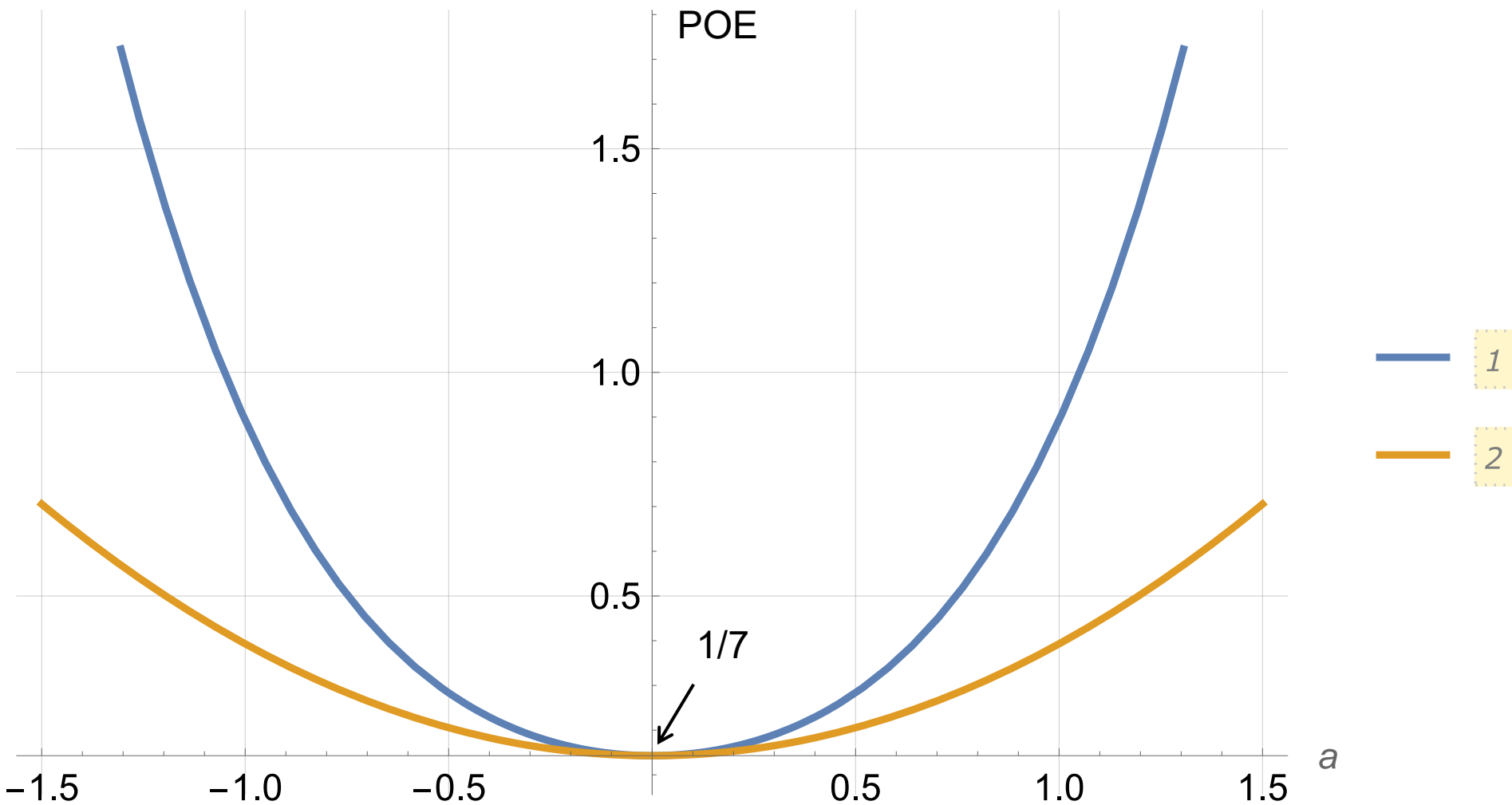
Let us suppose an experiment 2^2 with a **center point** replicated **3 times** ($N=7$) and model with **interaction**.

In the following the **POE** (Propagation of Error)

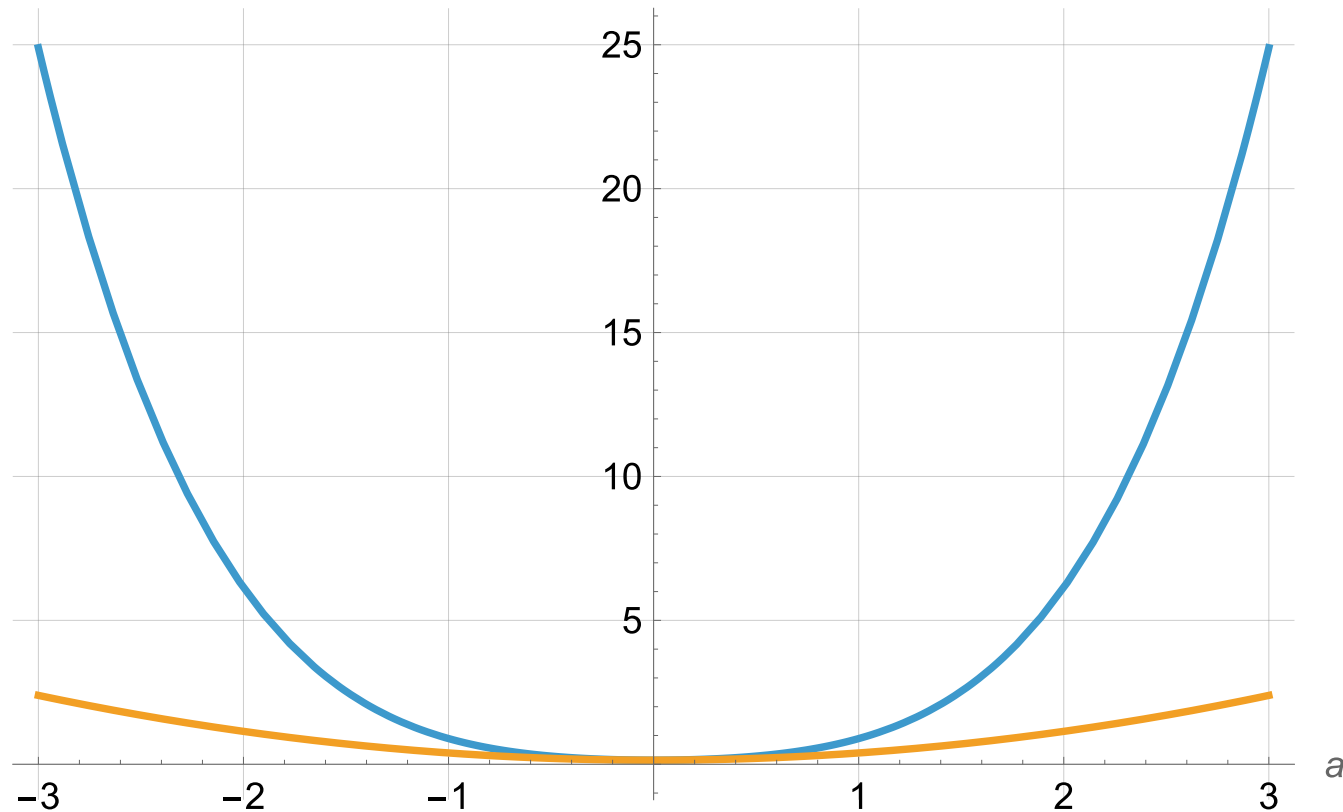
$$POE(\mathbf{x}) = \sqrt{\frac{V(\hat{y}(\mathbf{x}))}{\sigma^2}} = \sqrt{\mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}}$$



POE (Propagation of Error) for the Expected value:
1->Diagonal ($a=b$); **2-> along an Axis ($b=0$)**



If we extend the experimental range from $[-1.5, 1.5]$ to $[-3., 3.]$, we have



Prediction interval (**PI**) (for Point estimate)

Let us consider $y(\mathbf{x}) - \hat{y}(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\beta} + \varepsilon - \mathbf{x}^T \hat{\boldsymbol{\beta}}$

$$E(y(\mathbf{x}) - \hat{y}(\mathbf{x})) = E(\mathbf{x}^T \boldsymbol{\beta} + \varepsilon - \mathbf{x}^T \hat{\boldsymbol{\beta}}) = 0$$

$$V(y(\mathbf{x}) - \hat{y}(\mathbf{x})) = V(\mathbf{x}^T \boldsymbol{\beta} + \varepsilon - \mathbf{x}^T \hat{\boldsymbol{\beta}}) = V(\varepsilon) + V(\mathbf{x}^T \hat{\boldsymbol{\beta}})$$

$$= \sigma^2 + \mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x} \sigma^2$$

$$\frac{y(\mathbf{x}) - \hat{y}(\mathbf{x})}{\sqrt{MS_E [1 + \mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}]}} \sim t(N - p)$$

Note that the **PI** is larger than **CI**

Let us compute the **CI** for (8) (Example 10.1 -**Viscosity**)

Temperature = 96°C Speed = 10 lb/hr

$$\hat{y}_8 = 1566.08 + 7.6213 * 96 + 8.585 * 10 = 2383.57$$

$$\mathbf{x}_8 = [1 \ 96 \ 10] \quad \hat{\sigma}^2 = 266.9 \quad \mathbf{x}_8^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_8 = 0.0979706 \quad \sqrt{\mathbf{x}_8^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_8} = 0.31$$

$$\hat{y}_8 - t_{\alpha/2}(N-p) \sqrt{\hat{\sigma}^2 \mathbf{x}_8^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_8} \leq \mu_{y|\mathbf{x}_8} \leq \hat{y}_8 + t_{\alpha/2}(N-p) \sqrt{\hat{\sigma}^2 \mathbf{x}_8^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_8}$$

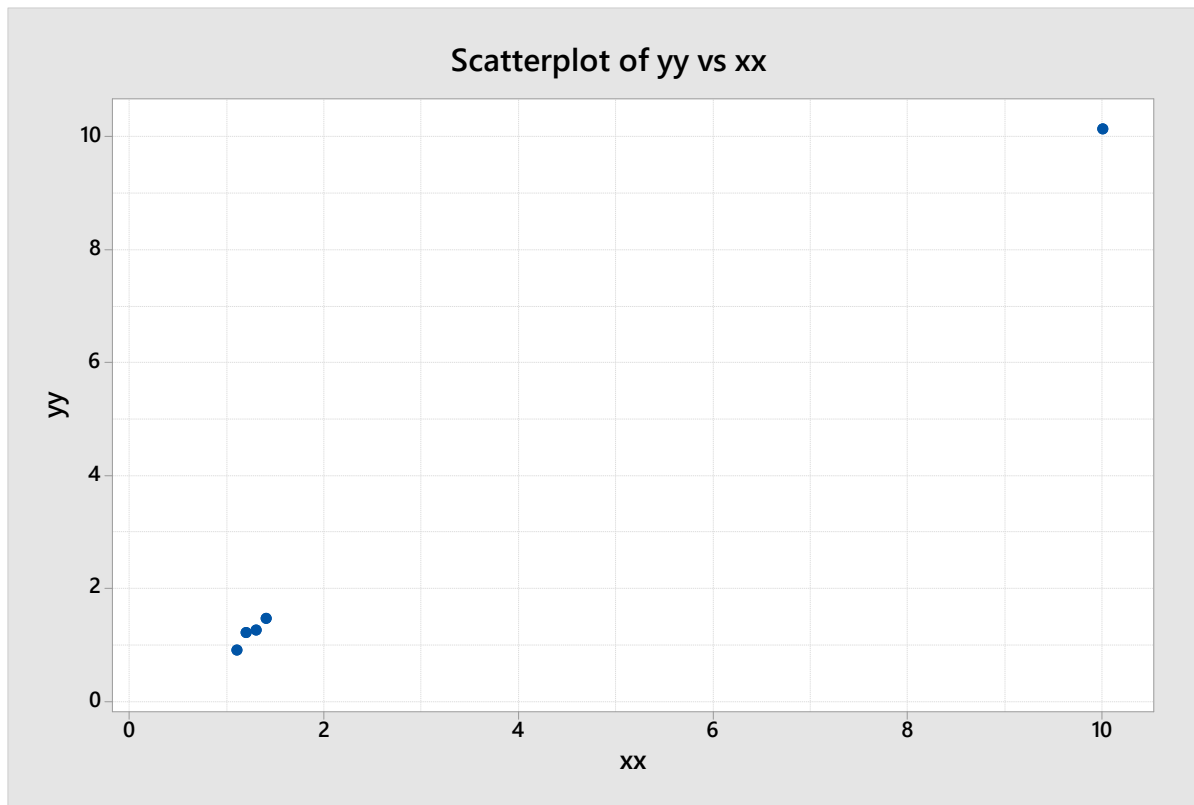
$$2383.57 - 2.16037 \sqrt{266.9 * 0.0979706} \leq \mu_{y|\mathbf{x}_8} \leq 2383.57 + 2.16037 \sqrt{266.9 * 0.0979706}$$

$$2383.57 - 2.16037 * 5.1135 \leq \mu_{y|\mathbf{x}_8} \leq 2383.57 + 2.16037 * 5.1135$$

$$2372.5 \leq \mu_{y|\mathbf{x}_8} \leq 2394.6$$

Warning: Do not use the model to predict too far from the experimental region

- Let us look at the previous yy-xx data



Example 1

yy	xx
0.90	1.1
1.20	1.2
1.25	1.3
1.45	1.4
10.12	10.0

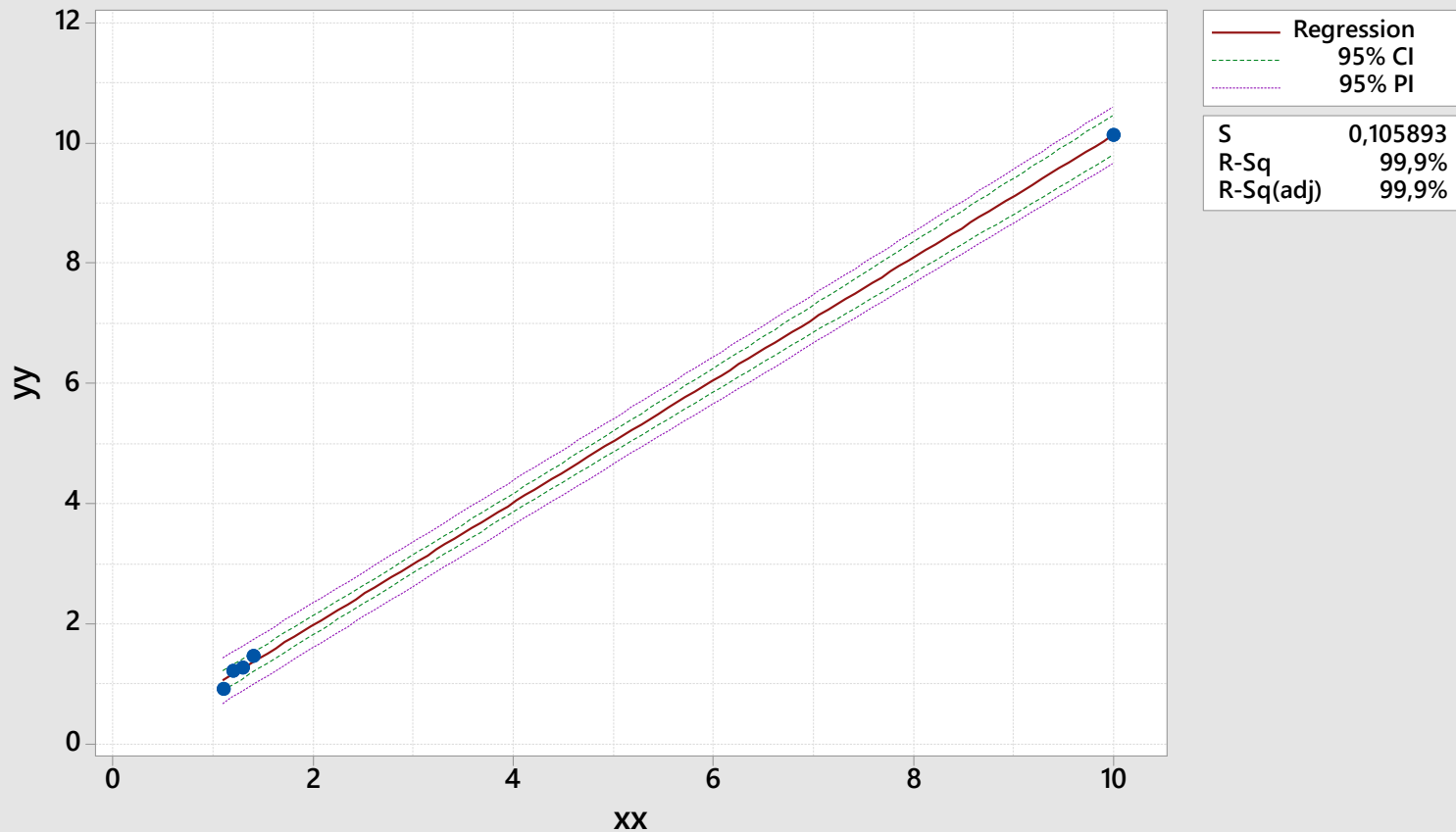
First order model (polynomial degree)

The regression equation is

$$yy = -0.07595 + 1.020 \ xx$$

S	R-sq	R-sq(adj)	R-sq(pred)
0.11	99.95%	99.93%	44.33%

Fitted Line Plot
 $yy = -0.07595 + 1.020 \ xx$



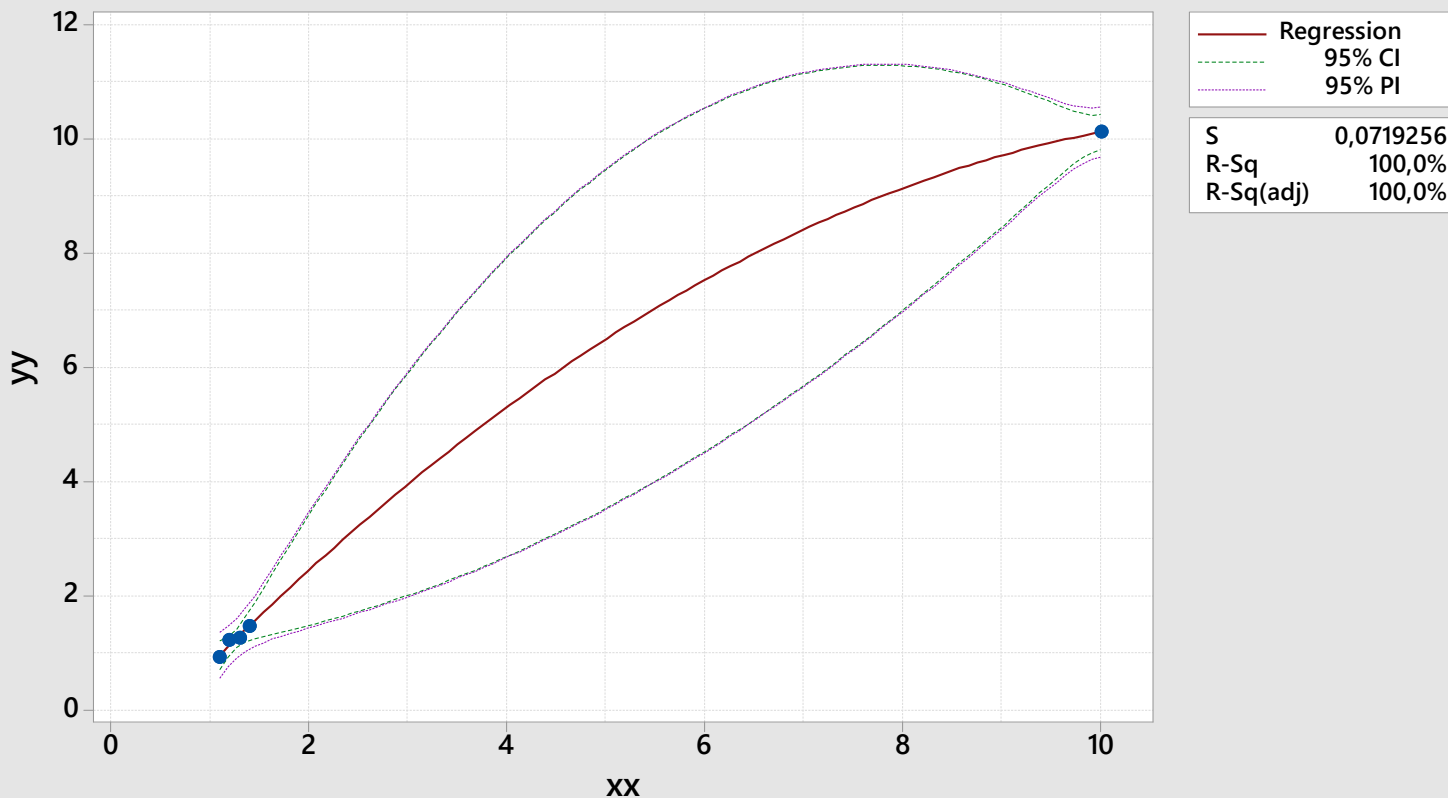
Second order model

The regression equation is

$$yy = - 1.049 + 1.897 \ xx - 0.07805 \ xx^2$$

S	R-sq	R-sq(adj)	R-sq(pred)
0.072	99.98%	99.97%	0.00%

Fitted Line Plot
 $yy = - 1,049 + 1,897 \ xx$
 $- 0,07805 \ xx^2$



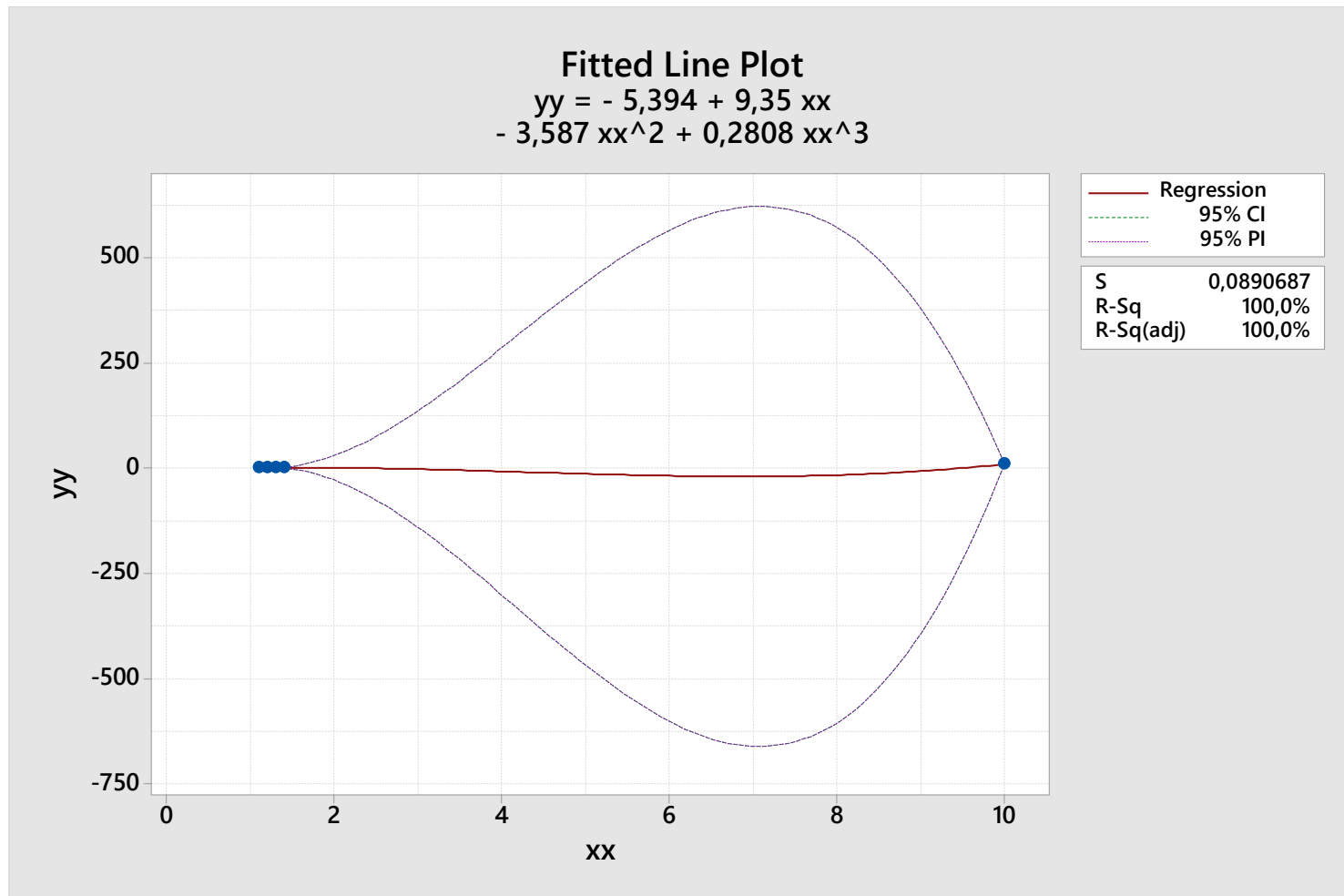
Third order model

The regression equation is

$$yy = - 5.394 + 9.35 \ xx - 3.587 \ xx^2 + 0.2808 \ xx^3$$

S	R-sq	R-sq(adj)	R-sq(pred)
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0.089	99.99%	99.95%	0.00%
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Von Neuman's elephant

Freeman Dyson reported a very interesting Enrico Fermi's statement in the paper «A meeting with Enrico Fermi» (Nature 2004)

I remember my friend Johnny von Neuman used to say: with four parameters I can fit an elephant, and with five I can make him wiggle the trunk!

And, among many papers on the same subject:

Dian JIN, Junze YUAN «Fitting an Elephant with four non-zero parameters» Arxiv 2024

<https://demonstrations.wolfram.com/FittingAnElephant/>

Confidence Bands of Expected response

When you are interested in **bands** around the response fits, you must consider that you are considering a set of points.

The formula* for the band around the **expected value (CI)** is

$$\text{Bands} = \mathbf{x}^T \hat{\boldsymbol{\beta}} \pm \sqrt{pF_{\alpha}(p, N-p)} \sqrt{MS_E \mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}}$$

The formula for the band around the **predicted individual value (PI)** is larger

$$\text{Bands} = \mathbf{x}^T \hat{\boldsymbol{\beta}} \pm \sqrt{pF_{\alpha}(p, N-p)} \sqrt{(1 + MS_E) \mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}}$$

Consider the previous **Example 1**, $N=5$, $p=2$ $\alpha=0.05$

$$\sqrt{pF_{0.05}(p, N-p)} = \sqrt{2 * 9.5521} = 4.37 \gg t_{0.025}(N-p) = t_{0.025}(3) = 2.228$$

*see Bates D.M. Watts D.G. «Non linear regression analysis and its applications» John Wiley 1988 pag. 6

Power of regression coefficients

$$y = \mathbf{x}^T \boldsymbol{\beta} + \varepsilon$$

$$\text{Under } H_0 : \beta_i = 0 \rightarrow \left\{ \begin{array}{l} \frac{\hat{\beta}_i}{\sqrt{MS_E (\mathbf{X}^T \mathbf{X})_{ii}^{-1}}} \sim t(df_E) \\ \frac{\hat{\beta}_i^2}{MS_E (\mathbf{X}^T \mathbf{X})_{ii}^{-1}} \sim F(1, df_E, \delta_i) \end{array} \right. \quad \leftarrow$$

$$\text{Under } H_1 : \beta_i = \beta_i^A \rightarrow \frac{\hat{\beta}_i^2}{MS_E (\mathbf{X}^T \mathbf{X})_{ii}^{-1}} \sim F(1, df_E, \delta_i)$$

$$\delta_i = \left(\frac{\beta_i^A}{\sigma} \right)^2 \frac{1}{(\mathbf{X}^T \mathbf{X})_{ii}^{-1}}$$

$$\Pr(F(1, df_E) > F_c) = \alpha \rightarrow \text{Power} = \Pr(F(1, df_E, \delta_{AB}) > F_c)$$

Poorly conditioned data

- Sometimes it is not possible to invert the $\mathbf{X}^T\mathbf{X}$ matrix. This happens when at least a column of the $\mathbf{X}$ matrix is a linear combination of other columns: **Exact multicollinearity**.
- Let us consider an experiment replicated two times with **one factor** and treatments at values 5, 6, 9
- Let us fit a polynomial of the **third order**

$$\mathbf{X} = \begin{pmatrix} 1 & 5 & 25 & 125 \\ 1 & 6 & 36 & 216 \\ 1 & 9 & 81 & 729 \\ 1 & 5 & 25 & 125 \\ 1 & 6 & 36 & 216 \\ 1 & 9 & 81 & 729 \end{pmatrix} \Rightarrow \text{Det}(\mathbf{X}^T \mathbf{X}) = 0$$

- The consequence is that we are not able to solve **Normal Equations** since they have infinite solutions.

2² experiment replicated

$$X = \begin{pmatrix} 1 & -1 & -1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix};$$

MatrixRank[X]->4

Transpose[X].X//MatrixForm

$$\begin{pmatrix} 8 & 0 & 0 & 0 & 8 & 8 \\ 0 & 8 & 0 & 0 & 0 & 0 \\ 0 & 0 & 8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 & 0 & 0 \\ 8 & 0 & 0 & 0 & 8 & 8 \\ 8 & 0 & 0 & 0 & 8 & 8 \end{pmatrix}$$

- But the projection onto the model space exists and it is unique!

- In some cases we have a **Near multicollinearity** (i.e. the determinant of the $\mathbf{X}^T\mathbf{X}$ matrix is close to zero).
- In these cases we say that the data are **poorly** or **ill** or **bad conditioned**.
- The consequence is that the variance of some model parameter is very high and the model will hardly be able to make good predictions.

VIF-Variance Inflation Factor

A nice method to look for **Multicollinearity** is to compute the **Variance Inflation Factor (VIF)** for each model parameter

For the Example 1

Coefficients model 1

Term	Coef	SE Coef	T-Value	P-Value	VIF
Const	-0.0760	0.0624	-1.22	0.310	
xx	1.0200	0.0135	75.41	0.000	1.00

Coefficients model 2

Term	Coef	SE Coef	T-Value	P-Value	VIF
Const	-1.049	0.460	-2.28	0.150	
xx	1.897	0.414	4.59	0.044	2026.67
xx*xx	-0.0780	0.0368	-2.12	0.168	2026.67

The **variance** of the model parameter j is

$$\hat{V}(\hat{\beta}_j) = \frac{MS_E}{(N-1)V(\mathbf{X}_j)} VIF_j$$

Where

$$VIF_j = \frac{1}{1 - R_j^2}$$

$\mathbf{X}_j$ is the j^{th} column of $\mathbf{X}$ matrix

R_j^2 is the coefficient of determination R^2 for the regression of $\mathbf{X}_j$ on the all other regressors.

N the total number of runs

Example 1 and a quadratic model: consider the additive term parameter

Regression Equation (to find VIF_1)

$$xx = 1.1097 + 0.08891 \, xx^2$$

$$VIF_1 = \frac{1}{1 - 0.9995} = 2026.67$$

Model Summary

S	R-sq	R-sq(adj)
0.100410	99.95%	99.93%

$$V(\hat{\beta}_1) = \frac{MS_E}{(N-1)V(X_1)} VIF_1 = \frac{0.0052}{(5-1)15.32} 2026.67 = 0.172$$

Try for the quadratic coefficient

Remark: the VIF coefficient depends on the model only

Proof

$$\hat{V}(\hat{\beta}_j) = MS_E \left[(\mathbf{X}^T \mathbf{X})^{-1} \right]_{jj}$$

$$\mathbf{r} = \mathbf{X}^T \mathbf{X}$$

Reorder columns and rows of $\mathbf{r}$ such that the first element is (j,j)

$$\mathbf{r}^{-1} = \begin{bmatrix} r_{j,j} & \mathbf{r}_{j,-j} \\ \mathbf{r}_{-j,j} & \mathbf{r}_{-j,-j} \end{bmatrix}^{-1} \quad \text{where}$$

$$r_{j,j} = \mathbf{X}_j^T \mathbf{X}_j \rightarrow \text{dim} : 1 \cdot 1$$

$$\mathbf{r}_{j,-j} = \mathbf{X}_j^T \mathbf{X}_{-j} \rightarrow \text{dim} : 1 \cdot k$$

$$\mathbf{r}_{-j,j} = \mathbf{X}_{-j}^T \mathbf{X}_j \rightarrow \text{dim} : k \cdot 1$$

$$\mathbf{r}_{-j,-j} = \mathbf{X}_{-j}^T \mathbf{X}_{-j} \rightarrow \text{dim} : k \cdot k$$

Let us use the Schur complement

$$r_{1,1}^{-1} = \left[r_{j,j} - \mathbf{r}_{j,-j} \mathbf{r}_{-j,-j}^{-1} \mathbf{r}_{-j,j} \right]^{-1}$$

Check element
dimensions

$$\begin{aligned} \hat{V}(\hat{\beta}_j) &= MS_E \left[\left(\mathbf{X}^T \mathbf{X} \right)^{-1} \right]_{j,j} = MS_E * r_{1,1}^{-1} \\ &= MS_E \left[\mathbf{X}_j^T \mathbf{X}_j - \mathbf{X}_j^T \mathbf{X}_{-j} \left(\mathbf{X}_{-j}^T \mathbf{X}_{-j} \right)^{-1} \mathbf{X}_{-j}^T \mathbf{X}_j \right]^{-1} \\ &= MS_E \left[\mathbf{X}_j^T \mathbf{X}_j - \mathbf{X}_j^T \mathbf{X}_{-j} \left(\mathbf{X}_{-j}^T \mathbf{X}_{-j} \right)^{-1} \left(\mathbf{X}_{-j}^T \mathbf{X}_{-j} \right) \left(\mathbf{X}_{-j}^T \mathbf{X}_{-j} \right)^{-1} \mathbf{X}_{-j}^T \mathbf{X}_j \right]^{-1} \\ &= MS_E \left[\mathbf{X}_j^T \mathbf{X}_j - \hat{\beta}_{*j}^T \left(\mathbf{X}_{-j}^T \mathbf{X}_{-j} \right) \hat{\beta}_{*j} \right]^{-1} \quad \hat{\beta}_{*j} = \left(\mathbf{X}_{-j}^T \mathbf{X}_{-j} \right)^{-1} \mathbf{X}_{-j}^T \mathbf{X}_j \\ &= MS_E \frac{1}{RSS_j} \\ &= \frac{MS_E}{(N-1)V(\mathbf{X}_j)} \frac{1}{1-R_j^2} = \frac{MS_E}{(N-1)V(\mathbf{X}_j)} VIF_j \end{aligned}$$

Conclusions

- Rule of thumb: **VIF** is critical if greater than **10**
- VIF depends on the experimental plan
- VIF depends on the model you use
- To avoid VIF (without re-designing the experiment) one could eliminate one or more predictors (**X** matrix columns)

Basis with orthogonal vectors

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \beta_0 \mathbf{X}_0 + \beta_1 \mathbf{X}_1 + \dots + \beta_k \mathbf{X}_k + \boldsymbol{\varepsilon}$$

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} \mathbf{X}_0^T \\ \mathbf{X}_1^T \\ \dots \\ \mathbf{X}_k^T \end{bmatrix} \begin{bmatrix} \mathbf{X}_0 & \mathbf{X}_1 & \dots & \mathbf{X}_k \end{bmatrix} = \begin{bmatrix} \mathbf{X}_0^T \mathbf{X}_0 & 0 & \dots & 0 \\ 0 & \mathbf{X}_1^T \mathbf{X}_1 & \dots & 0 \\ \dots & \dots & \dots & 0 \\ 0 & 0 & \dots & \mathbf{X}_k^T \mathbf{X}_k \end{bmatrix}$$

$$(\mathbf{X}^T \mathbf{X})^{-1} = \begin{bmatrix} 1 / \mathbf{X}_0^T \mathbf{X}_0 & \dots & 0 \\ \dots & \dots & \dots \\ 0 & \dots & 1 / \mathbf{X}_k^T \mathbf{X}_k \end{bmatrix}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \begin{bmatrix} \mathbf{X}_0^T \mathbf{y} / \mathbf{X}_0^T \mathbf{X}_0 \\ \mathbf{X}_1^T \mathbf{y} / \mathbf{X}_1^T \mathbf{X}_1 \\ \dots \\ \mathbf{X}_k^T \mathbf{y} / \mathbf{X}_k^T \mathbf{X}_k \end{bmatrix} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \dots \\ \hat{\beta}_k \end{bmatrix}$$

$$\begin{aligned}
 SS_{REG}^* &= \hat{\boldsymbol{\beta}}^T \mathbf{X}^T \mathbf{y} = \hat{\beta}_0 \mathbf{X}_0^T \mathbf{y} + \hat{\beta}_1 \mathbf{X}_1^T \mathbf{y} + \dots + \hat{\beta}_k \mathbf{X}_k^T \mathbf{y} = \\
 &= \sum_{i=0,k} \hat{\beta}_i \mathbf{X}_i^T \mathbf{y} = \sum_{i=0,k} \frac{\mathbf{X}_i^T \mathbf{y}}{\mathbf{X}_i^T \mathbf{X}_i} \mathbf{X}_i^T \mathbf{y} = \sum_{i=0,k} \frac{\mathbf{y}^T \mathbf{X}_i \mathbf{X}_i^T \mathbf{y}}{\mathbf{X}_i^T \mathbf{X}_i}
 \end{aligned}$$

- Note we could **orthogonalize** the $\mathbf{X}$ matrix to simplify the analysis
- From a geometrical point of view this means to change the vector basis used to span the Model space.
- The **projections** will have a different form but their length (and SS, their squared length) will be the same

Trasformations

- Why transformations?
 - Simplest model
 - Variance Homogeneity (but we could use WLS)
 - Normality hypothesis
- Response Variable transformation
- Predictor transformation or elimination
 - Predictor coding is always useful
 - Empirical rule: The model structure should not change after rigid translation or rotation of the predictor reference system

Y (response variable) transformation

When: $y_{\max}/y_{\min} > 10$

We could use a power transformation y^λ

Let us assume that the model is **full quadratic**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2 + \varepsilon$$

Minimize one of

a) $SS(\hat{\beta}_{11}, \hat{\beta}_{22}, \hat{\beta}_{12} \mid \hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2)$

b) MS_1 / MS_2

c) SS_E (Box-Cox)

Polynomial models (in one factor)

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \dots + \beta_k x_i^k + \varepsilon_i$$

- When we change the polynomial order, the parameter estimates change if the design is not orthogonal
- We have to simplify the model, step by step (there are automatic procedure called **Stepwise** or **Best subset** or...)
- There are two general common sense criteria :
 - ✓ After the simplification, the **functional form** of the Model should **not** change when you apply a **rigid translation of the axes**
 - ✓ After the simplification, the **functional form** of the Model should **not** change when you apply a **rigid rotation of the axes** (This does not apply in one factor case)

Example 1 (rigid translation)

- Assume $y = \beta_0 + \beta_1 x + \beta_{11} x^2 + \varepsilon$
- Apply a **rigid translation** of axes $x = z + a$
$$E(y) = \alpha_0 + \alpha_1 z + \alpha_{11} z^2$$
$$\alpha_0 = \beta_0 + \beta_1 a + \beta_{11} a^2$$
 - With $\alpha_1 = \beta_1 + 2\beta_{11} a$
$$\alpha_{11} = \beta_{11}$$
- If we eliminate the linear term β_1 , the functional form of the Model has still the linear term **after** any rigid traslation: **do not make that**

The same is true for more complex polynomials

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \varepsilon$$

Apply a **rigid translation** of axes

$$x_1 = z_1 + a_1 \quad x_2 = z_2 + a_2$$

$$y = \alpha_0 + \alpha_1 z_1 + \alpha_2 z_2 + \alpha_{12} z_1 z_2 + \alpha_{11} z_1^2 + \alpha_{22} z_2^2 + \varepsilon$$

$$\alpha_0 = \beta_0 + \beta_1 a_1 + \beta_2 a_2 + \beta_{12} a_1 a_2 + \beta_{11} a_1^2 + \beta_{22} a_2^2$$

$$\alpha_1 = \beta_1 + 2\beta_{11} a_1 + \beta_{12} a_2$$

$$\alpha_2 = \beta_2 + 2\beta_{22} a_2 + \beta_{12} a_1$$

$$\alpha_{12} = \beta_{12}$$

$$\alpha_{11} = \beta_{11}$$

$$\alpha_{22} = \beta_{22}$$

i.e. If we eliminate **beta**₁ after a rigid translation, **alfa**₁ will appear in the new model

Example 2 (Rigid rotation)

- Assume $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \varepsilon$

$$y = \beta_0 + \mathbf{x}^T \boldsymbol{\beta} + \mathbf{x}^T \mathbf{B} \mathbf{x} + \varepsilon$$

- Apply a rigid rotation of axes, with angle θ

$$\mathbf{z} = \mathbf{M} \mathbf{x} \quad \text{con } \mathbf{M} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \quad \text{and } \mathbf{M}^{-1} = \mathbf{M}^T$$

$$\mathbf{x} = \mathbf{M}^{-1} \mathbf{z} = \mathbf{M}^T \mathbf{z}$$

$$y = \beta_0 + \mathbf{x}^T \boldsymbol{\beta} + \mathbf{x}^T \mathbf{B} \mathbf{x} + \varepsilon$$

$$\mathbf{x} = \mathbf{M}^{-1} \mathbf{z} = \mathbf{M}^T \mathbf{z}$$

$$\mathbf{x}^T \boldsymbol{\beta} = \mathbf{z}^T \mathbf{M} \boldsymbol{\beta} = \mathbf{z}^T \boldsymbol{\alpha}$$

$$\mathbf{x}^T \mathbf{B} \mathbf{x} = \mathbf{z}^T \mathbf{M} \mathbf{B} \mathbf{M}^T \mathbf{z} = \mathbf{z}^T \mathbf{A} \mathbf{z}$$

$$\alpha_0 = \beta_0$$

$$\alpha_1 = c\beta_1 + s\beta_2$$

$$\alpha_2 = -s\beta_1 + c\beta_2$$

$$\alpha_{12} = -2sc\beta_{11} + 2sc\beta_{22} + (c^2 - s^2)\beta_{12}$$

$$\alpha_{11} = c^2\beta_{11} + s^2\beta_{22} + sc\beta_{12}$$

$$\alpha_{22} = s^2\beta_{11} + c^2\beta_{22} - sc\beta_{12}$$

$$y = \alpha_0 + \mathbf{z}^T \boldsymbol{\alpha} + \mathbf{z}^T \mathbf{A} \mathbf{z} + \varepsilon$$

You can see immediately the consequences of eliminating some term from the original model

- Summing up: two practical rules:
 - **Rule 1**: If one wants that the model **functional form** is the same after a **rigid translation of axes**, eliminate in each step the polynomial term with the **maximum degree** until the test fails to reject.
 - **Rule 2**: : If one wants that the model **functional form** is the same after a **rigid rotation of axes**, eliminate in each step **all** polynomial terms with the **maximum degree** until the test fails to reject.

Orthogonal Polynomials

- We could use the orthogonal polynomials

$$y_i = \beta_0 + \beta_1 \varphi_1(x_i) + \beta_2 \varphi_2(x_i) + \dots + \beta_k \varphi_k(x_i) + \varepsilon_i$$

$$\sum_{i=1, N} \varphi_l(x_i) \varphi_m(x_i) = 0 \quad l \neq m$$

- We can simplify the model in just one step applying the F test to each term simultaneously (perhaps using the Bonferroni approach)
- In Classwork you will see a numerical application

Effect on the optimization

- Let us suppose that the true function is
$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon \quad \text{with} \quad \varepsilon \sim N(0, \sigma^2)$$

- You make your experiment and you fit the function

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2 \quad \text{and} \quad \Sigma_{\hat{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \hat{\sigma}^2 \quad \text{and} \quad \hat{\sigma}^2 \equiv MS_E$$

- Let us suppose that you want to find the unconstrained minimum of the expected value of the response variable

$$E(y) \quad \longrightarrow \quad E(\hat{y})$$

- Usually you apply the N&S conditions to find the minimum

$$\frac{d\hat{y}}{dx} = 0 \text{ and } \frac{d^2\hat{y}}{dx^2} > 0$$

- After some algebraic manipulation

$$x_{opt} = -\frac{\hat{\beta}_1}{2\hat{\beta}_2} \text{ and } \hat{\beta}_2 > 0$$

$$\Sigma_{\hat{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \hat{\sigma}^2$$

$$y_{opt} = \hat{\beta}_0 - \frac{\hat{\beta}_1^2}{4\hat{\beta}_2}$$

- The problem is to find an interval R_0 such that

$$\Pr[x_{opt} \in R_0] \geq 1 - \alpha$$

Methods

- Analytical
- Monte Carlo

Analytical Method

Consider the estimated model parameters excluding the model constant

$$E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\mu}_{\hat{\boldsymbol{\beta}}} = \begin{Bmatrix} \mu_1 \\ \mu_2 \end{Bmatrix} \quad \boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} = (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2$$

Developing in Taylor series x_{opt} to the **fourth** term (the expansion to the eighth term changes the figures only to the fourth significant figure) we have

Expected value and Variance

$$E\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right) = \frac{-\mu_1\mu_2^4 + \rho\mu_2^3\sigma_1\sigma_2 - \mu_1\mu_2^2\sigma_2^2 + 3\rho\mu_2\sigma_1\sigma_2^3 - 3\mu_1\sigma_2^4}{2\mu_2^5}$$

$$V\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right) = \frac{1}{4\mu_2^{10}}(\mu_2^8\sigma_1^2 - 2\rho\mu_1\mu_2^7\sigma_1\sigma_2 + \mu_1^2\mu_2^6\sigma_2^2 + 3\mu_2^6\sigma_1^2\sigma_2^2 + 5\rho^2\mu_2^6\sigma_1^2\sigma_2^2 - 16\rho\mu_1\mu_2^5\sigma_1\sigma_2^3 + \\ + 8\mu_1^2\mu_2^4\sigma_2^4 + 15\mu_2^4\sigma_1^2\sigma_2^4 + 54\rho^2\mu_2^4\sigma_1^2\sigma_2^4 - 108\rho\mu_1\mu_2^3\sigma_1\sigma_2^5 + 39\mu_1^2\mu_2^2\sigma_2^6 + \\ + 45\mu_2^2\sigma_1^2\sigma_2^6 + 261\rho^2\mu_2^2\sigma_1^2\sigma_2^6 - 402\rho\mu_1\mu_2\sigma_1\sigma_2^7 + 96\mu_1^2\sigma_2^8 + 105\sigma_1^2\sigma_2^8 + 840\rho^2\sigma_1^2\sigma_2^8)$$

Assuming Gaussian distribution and known variance

$$R_0 = \left[E\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right) - z_{\alpha/2} \sqrt{V\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right)}; \quad E\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right) + z_{\alpha/2} \sqrt{V\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right)} \right]$$

Monte Carlo

Let us suppose that the true model is

$$y = 4 - 5x + x^2 + \varepsilon \quad \text{with} \quad \varepsilon \sim N(0, 3^2)$$

Consider:

- the experimentation treatments are 12,
- the experimentation domain is $[-3, 8]$
- the treatments are equidistant $[-3, -2, \dots, 7, 8]$
- No replications
- Monte Carlo with 10000 runs

We generate the **Histogram** of the optima and we pick up the quantiles 2.5% and 97.5%. We have

$$R_0=[2.318;2.827]$$

If we use the **Analytical solution** we have

$$R_0=[2.317; 2.827]$$

The true optimum is 2.5

Different stochastic problems

$$\text{Min}_x \hat{y}(x)$$

$$\text{Min}_x \hat{y}(x)$$

s.t $x \in [a, b]$ a and b deterministic constant

$$\text{Min}_x E\left(y\left(x, \hat{T}(x)\right)\right)$$

$$y = \frac{\alpha}{v} \left(1 + \frac{\varphi}{\hat{T}(v)} \right)$$

$$\text{s.t } \Pr\left\{\hat{P}(x) \leq P_0\right\} = 1 - \alpha$$

$$\hat{P}(v) = \eta \hat{F} v$$

And many other problems